

R489: Outer Bridges II

TFPT RH programme

31 August 2026

Claim boundary. This is a formalization and gap-reduction round. It proves two continuity statements and leaves one typed dense-completion package. It is not an RH claim.

1 Topology

The r487 carrier is corrected to use an explicit compactly supported $L^2(\mathbb{R})$ witness h , rather than merely L^1 , with

$$g(u) = \int_{\mathbb{R}} h(t)h(t+u) dt.$$

For a target supported in $[-R, R]$, a grid sequence g_j now satisfies:

1. $\text{supp } g_j \subseteq [-R, R]$;
2. $g_j \rightarrow g$ uniformly on $[-R, R]$;
3. every g_j is interval-integrable there;
4. $\int_{-R}^R |g_j - g| \rightarrow 0$.

Uniform convergence is the right input for the finite set of prime point evaluations. The explicit L^1 component is retained because it gives bounded-weight integral continuity without deriving it again from uniform convergence and interval length.

2 Prime channel: closed

Define the common support anchor

$$A_R = \max\{1, \lceil e^R \rceil\}.$$

Lean proves $\text{elementAnchor}(g_j) \leq A_R$, hence every native prime side is the same finite sum

$$\sum_{n \in \text{windowAtoms}(A_R)} \frac{2\Lambda(n)}{\sqrt{n}} g_j(\log n).$$

Global pointwise convergence follows from uniform convergence on the support and simultaneous vanishing outside it. The theorem `fullWeilCombSide_tendsto` then follows from `tendsto_finset_sum`. This channel is sorry-free.

3 Polar channel: continuity closed, dictionary exposed

The standard compact-support polar functional is now concrete:

$$P(g) = \int_{-R}^R -2 \cosh(u/2) g(u) du.$$

On $[-R, R]$, its weight is bounded by $2 \cosh(R/2)$. Lean proves

$$|P(g_j) - P(g)| \leq 2 \cosh(R/2) \int_{-R}^R |g_j(u) - g(u)| du \longrightarrow 0$$

as `fullWeilPoleIntegral_tendsto`.

The internal `r376` object `weilPoleSide` is, however, still the native-mesh second-difference pairing of `polePotential`. Its finite-element equality with $P(g_j)$ is not definitional and was already represented by the unproved mesh dictionary in `r376`. `R489` names its exact landing site `GridPoleIntegralIdentification`; no equality is silently assumed.

4 Single remaining completion

The dense bridge now has one direct sorry, `fullWeil_fixedSupport_completion`, whose proposition is the transparent conjunction:

1. `FullWeilFixedSupportGridDensity`: dyadic step approximation of compactly supported L^2 witnesses and uniform convergence of their autocorrelations;
2. `FullWeilArchContinuity`: continuity of `r475`'s concrete u -space archimedean pairing;
3. `GridPoleIntegralIdentification`: the `r376` second-difference/integral finite-element dictionary.

The public density and channel-continuity theorems are projections and compositions of this single package; the prime and standard polar continuity subproofs are already closed.

For the arch component, Gauss–digamma is not required merely to prove continuity of the concrete `r475` u -space pairing. It remains needed when that pairing is identified with the standard spectral Guinand–Weil normalization in the separate explicit-formula bridge.

5 Mathlib status and census

Mathlib supplies L^p spaces, Hölder inequalities, Bochner integration, interval-integral bounds, simple functions, and group convolution. No directly reusable theorem was found that simultaneously produces compactly supported dyadic steps and proves uniform convergence of translated autocorrelations. The intended next proof is

$$\|h_j \star \tilde{h}_j - h \star \tilde{h}\|_\infty \leq (\|h_j\|_2 + \|h\|_2) \|h_j - h\|_2,$$

after a support-preserving dyadic L^2 -density lemma.

The external sorry census decreases from four to three and the repository census from 9 to 8: the two `r487` density/channel sorries are replaced by one completion sorry. The other external sorries remain explicit-formula normalization and off-critical separation.